

14.1 MEANS AND COVARIANCES OF NONLINEAR TRANSFORMATIONS

In this section, we will show how linearization approximations can result in errors in the transformation of means and covariances when a random variable is operated on by a nonlinear function. This section does not really have anything to do directly with state estimation, Kalman filtering, or the UKF. However, this section provides some background that will allow us to develop the UKF later in this chapter. This section will also give us a more complete background to understand the type of problems that can arise in the EKF (which relies on linearization).

Consider the nonlinear transformation

$$\begin{aligned} y_1 &= r \cos \theta \\ y_2 &= r \sin \theta \end{aligned} \quad (14.1)$$

This is a standard polar-to-rectangular coordinate transformation. For instance, we might have a sensor that measures range r and angle θ , and we want to convert the measured data to rectangular coordinates y_1 and y_2 . The coordinate transformation can be written more generally as

$$y = h(x) \quad (14.2)$$

where y is the two-element output of $h(x)$, and the two-element vector x is defined as

$$x = \begin{bmatrix} r \\ \theta \end{bmatrix} \quad (14.3)$$

Suppose that x_1 (which is the range r) is a random variable with a mean of 1 and a standard deviation of σ_r . Suppose that x_2 (which is the angle θ) is a random variable with a mean of $\pi/2$ and a standard deviation of σ_θ . In other words, the means of the components of x are given as $\bar{r} = 1$ and $\bar{\theta} = \pi/2$. In addition, we will assume that r and θ are independent, and that their probability density functions are symmetric around their means (for example, Gaussian or uniform).

14.1.1 The mean of a nonlinear transformation

An initial consideration of the above problem, along with Equation (14.1), would lead us to believe that y_1 has a mean of 0, and y_2 has a mean of 1. In addition, a linearization approach would lead us to the same conclusion. If we perform a first-order linearization of Equation (14.2) and take the expected value of both sides, we obtain

$$\begin{aligned} \bar{y} &= E[h(x)] \\ &\approx E \left[h(\bar{x}) + \frac{\partial h}{\partial x} \bigg|_{\bar{x}} (x - \bar{x}) \right] \\ &= h(\bar{x}) + \frac{\partial h}{\partial x} \bigg|_{\bar{x}} E(x - \bar{x}) \\ &= h(\bar{x}) \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned} \quad (14.4)$$

Our intuition, along with a first-order linearization analysis, both lead us to the same conclusion. However, let us pursue this problem with more rigor to check our previous analysis. We can write r and θ as

$$\begin{aligned} r &= \bar{r} + \tilde{r} \\ \theta &= \bar{\theta} + \tilde{\theta} \end{aligned} \quad (14.5)$$

where $\tilde{r}$ and $\tilde{\theta}$ are simply the deviations of r and θ from their means. A rigorous analysis of the mean of y_1 can be performed as follows:

$$\begin{aligned} \bar{y}_1 &= E(r \cos \theta) \\ &= E[(\bar{r} + \tilde{r}) \cos(\bar{\theta} + \tilde{\theta})] \\ &= E[(\bar{r} + \tilde{r})(\cos \bar{\theta} \cos \tilde{\theta} - \sin \bar{\theta} \sin \tilde{\theta})] \end{aligned} \quad (14.6)$$

Carrying out the multiplication, remembering that $\tilde{r}$ and $\tilde{\theta}$ are independent with symmetric pdfs, and taking the expected value, results in

$$\begin{aligned} \bar{y}_1 &= \bar{r} \cos \bar{\theta} \\ &= 0 \end{aligned} \quad (14.7)$$

Our intuition and our first-order approximation of $\bar{y}_1$ have been confirmed by rigorous analysis. Let us repeat the process for y_2 :

$$\begin{aligned} \bar{y}_2 &= E(r \sin \theta) \\ &= E[(\bar{r} + \tilde{r}) \sin(\bar{\theta} + \tilde{\theta})] \\ &= E[(\bar{r} + \tilde{r})(\sin \bar{\theta} \cos \tilde{\theta} + \cos \bar{\theta} \sin \tilde{\theta})] \end{aligned} \quad (14.8)$$

Carrying out the multiplication, remembering that $\tilde{r}$ and $\tilde{\theta}$ are independent with symmetric pdfs, and taking the expected value, results in

$$\begin{aligned} \bar{y}_2 &= \bar{r} \sin \bar{\theta} E(\cos \tilde{\theta}) \\ &= E(\cos \tilde{\theta}) \end{aligned} \quad (14.9)$$

We cannot go any further unless we assume some distribution for $\tilde{\theta}$, so let us assume that $\tilde{\theta}$ is uniformly distributed between $\pm\theta_m$. In that case, we can compute

$$\begin{aligned} \bar{y}_2 &= E(\cos \tilde{\theta}) \\ &= \frac{\sin \theta_m}{\theta_m} \end{aligned} \quad (14.10)$$

We expected to get 1 for our answer in confirmation of Equation (14.4), but instead we got some number that is less than 1. [Note that $(\sin \theta_m)/\theta_m < 1$ for all $\theta_m > 0$, and $\lim_{\theta_m \rightarrow 0} (\sin \theta_m)/\theta_m = 1$.] The analysis reveals a problem with our initial intuition and the first-order linearization that we performed earlier. The mean of y_2 will indeed be less than 1. This can be seen by looking at a plot of 300 randomly generated r and θ values, where $\tilde{r}$ is uniformly distributed between ± 0.01 , and $\tilde{\theta}$ is uniformly distributed between ± 0.35 radians. The small variance of $\tilde{r}$ and the large

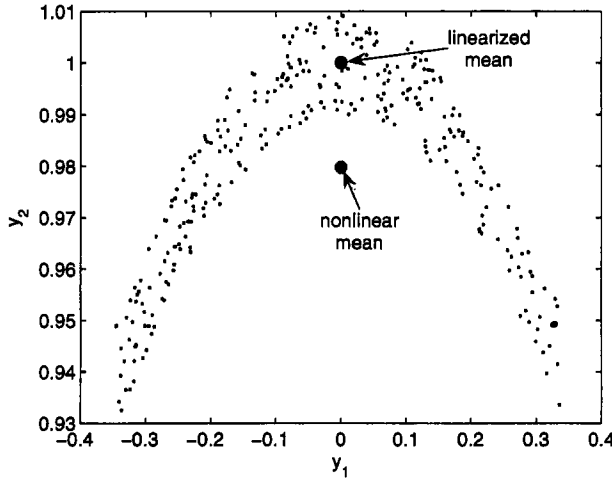


Figure 14.1 Linearized and nonlinear mean of 300 randomly generated points with $\tilde{r}$ uniformly distributed between ± 0.01 and $\tilde{\theta}$ uniformly distributed between ± 0.35 radians.

variance of $\tilde{\theta}$ result in an arc-shaped distribution of points as seen in Figure 14.1. This arc-shaped distribution results in $\bar{y}_2 < 1$.

This is not a Kalman filtering example. But since the EKF uses first-order linearization to update the mean of the state, this example shows the kind of error that can creep into the EKF when it is applied to a nonlinear system.

For a more general analysis of the mean of a nonlinear transformation, recall from Equation (1.89) that $y = h(x)$ can be expanded in a Taylor series around $\bar{x}$ as follows:

$$\begin{aligned} y &= h(x) \\ &= h(\bar{x}) + D_{\bar{x}}h + \frac{1}{2!}D_{\bar{x}}^2h + \frac{1}{3!}D_{\bar{x}}^3h + \dots \end{aligned} \quad (14.11)$$

where $\tilde{x} = x - \bar{x}$. The mean of y can therefore be expanded as

$$\begin{aligned} \bar{y} &= E \left[h(\bar{x}) + D_{\bar{x}}h + \frac{1}{2!}D_{\bar{x}}^2h + \frac{1}{3!}D_{\bar{x}}^3h + \dots \right] \\ &= h(\bar{x}) + E \left[D_{\bar{x}}h + \frac{1}{2!}D_{\bar{x}}^2h + \frac{1}{3!}D_{\bar{x}}^3h + \dots \right] \end{aligned} \quad (14.12)$$

By using $D_{\bar{x}}h$ from Equation (1.88) we can see that

$$\begin{aligned} E[D_{\bar{x}}h] &= E \left[\sum_{i=1}^n \tilde{x}_i \frac{\partial}{\partial x_i} h(x) \Big|_{x=\bar{x}} \right] \\ &= \sum_{i=1}^n E(\tilde{x}_i) \frac{\partial}{\partial x_i} h(x) \Big|_{x=\bar{x}} \\ &= 0 \end{aligned} \quad (14.13)$$

because $E(\tilde{x}_i) = 0$. Likewise, we can see that

$$\begin{aligned} E[D_{\tilde{x}}^3 h] &= E \left[\left(\sum_{i=1}^n \tilde{x}_i \frac{\partial}{\partial x_i} \right)^3 h(x) \Big|_{x=\tilde{x}} \right] \\ &= 0 \end{aligned} \quad (14.14)$$

This is because the sum in the above equation consists only of third-order moments [$E(\tilde{x}_1^3)$, $E(\tilde{x}_1^2 \tilde{x}_2)$, etc.]. These expected values will always be zero as shown at the end of Section 2.2. Similarly all of the odd terms in Equation (14.12) will be zero, which leads to the simplification

$$\bar{y} = h(\bar{x}) + \frac{1}{2!} E[D_{\tilde{x}}^2 h] + \frac{1}{4!} E[D_{\tilde{x}}^4 h] + \dots \quad (14.15)$$

This shows why the mean calculation in Equation (14.4) was incorrect; that calculation was only correct up to the first order. If we approximate $\bar{y}$ for our polar-to-rectangular transformation using terms up to the second order from Equation (14.15), we obtain

$$\begin{aligned} \bar{y} &\approx h(\bar{x}) + \frac{1}{2!} E[D_{\tilde{x}}^2 h] \\ &= h(\bar{x}) + \frac{1}{2} E \left[\left(\sum_{i=1}^2 \tilde{x}_i \frac{\partial}{\partial x_i} \right)^2 h(x) \Big|_{x=\tilde{x}} \right] \\ &= h(\bar{x}) + \frac{1}{2} \left(E(\tilde{x}_1^2) \frac{\partial^2 h(x)}{\partial x_1^2} \Big|_{x=\tilde{x}} + 2E(\tilde{x}_1 \tilde{x}_2) \frac{\partial^2 h(x)}{\partial x_1 \partial x_2} \Big|_{x=\tilde{x}} + E(\tilde{x}_2^2) \frac{\partial^2 h(x)}{\partial x_2^2} \Big|_{x=\tilde{x}} \right) \\ &= h(\bar{x}) + \frac{1}{2} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \sigma_\theta^2 \begin{bmatrix} -r \cos \theta \\ -r \sin \theta \end{bmatrix} \Big|_{x=\tilde{x}} \right) \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{1}{2} \sigma_\theta^2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} \end{aligned} \quad (14.16)$$

We therefore obtain

$$\begin{aligned} \bar{y}_1 &\approx 0 \\ \bar{y}_2 &\approx 1 - \frac{\sigma_\theta^2}{2} \\ &= 1 - \frac{E(\tilde{\theta}^2)}{2} \end{aligned} \quad (14.17)$$

Note that we found the exact value of $\bar{y}_2$ in Equation (14.9) to be equal to $E(\cos \tilde{\theta})$. The approximate expression found in Equation (14.17) is the first two nonzero terms of the Taylor series expansion of $E(\cos \tilde{\theta})$.

14.1.2 The covariance of a nonlinear transformation

Now we turn our attention to the covariance of a random variable that undergoes a nonlinear transformation. The covariance of y is given as

$$P_y = E [(y - \bar{y})(y - \bar{y})^T] \quad (14.18)$$

We can use Equations (14.11) and (14.15) to write $(y - \bar{y})$ as

$$\begin{aligned} y - \bar{y} &= \left[h(\bar{x}) + D_{\bar{x}}h + \frac{1}{2!}D_{\bar{x}}^2h + \dots \right] - \\ &\quad \left[h(\bar{x}) + \frac{1}{2!}E(D_{\bar{x}}^2h) + \frac{1}{4!}E(D_{\bar{x}}^4h) + \dots \right] \\ &= \left[D_{\bar{x}}h + \frac{1}{2!}D_{\bar{x}}^2h + \dots \right] - \left[\frac{1}{2!}E(D_{\bar{x}}^2h) + \frac{1}{4!}E(D_{\bar{x}}^4h) + \dots \right] \end{aligned} \quad (14.19)$$

We substitute this expression into Equation (14.18) and use the same type of reasoning as in the previous section to see that all of the odd-powered terms in the expected value evaluate to zero (assuming that $\bar{x}$ is zero-mean with a symmetric pdf). This results in

$$\begin{aligned} P_y &= E[D_{\bar{x}}h(D_{\bar{x}}h)^T] + \\ &\quad E\left[\frac{D_{\bar{x}}h(D_{\bar{x}}^3h)^T}{3!} + \frac{D_{\bar{x}}^2h(D_{\bar{x}}^2h)^T}{2!2!} + \frac{D_{\bar{x}}^3h(D_{\bar{x}}h)^T}{3!}\right] + \\ &\quad E\left(\frac{D_{\bar{x}}^2h}{2!}\right)E\left(\frac{D_{\bar{x}}^2h}{2!}\right)^T + \dots \end{aligned} \quad (14.20)$$

The first term on the right side of the above equation can be written as

$$\begin{aligned} E[D_{\bar{x}}h(D_{\bar{x}}h)^T] &= E\left[\left(\sum_{i=1}^n \tilde{x}_i \frac{\partial h}{\partial x_i} \Big|_{x=\bar{x}}\right) \left(\dots\right)^T\right] \\ &= E\left[\sum_{i,j} \tilde{x}_i \frac{\partial h}{\partial x_i} \Big|_{x=\bar{x}} \frac{\partial h^T}{\partial x_j} \Big|_{x=\bar{x}} \tilde{x}_j\right] \\ &= \sum_{i,j} H_i E(\tilde{x}_i \tilde{x}_j) H_j^T \\ &= \sum_{i,j} H_i P_{ij} H_j^T \end{aligned} \quad (14.21)$$

where the partial derivative vector H_i and the expected value P_{ij} are defined by the above equation. Recall from Equation (1.16) that an equation in this form can be written as

$$\begin{aligned} E[D_{\bar{x}}h(D_{\bar{x}}h)^T] &= \frac{\partial h}{\partial x} \Big|_{x=\bar{x}} P \frac{\partial h^T}{\partial x} \Big|_{x=\bar{x}} \\ &= HPH^T \end{aligned} \quad (14.22)$$

where the partial derivative matrix H and the covariance matrix P are defined by the above equation. H_i in Equation (14.21) is the i th column of H , and P_{ij} in Equation (14.21) is the element in the i th row and j th column of $P = E(\bar{x}\bar{x}^T)$. We can use this in Equation (14.20) to write the covariance of a nonlinear transformation $y = h(x)$ as follows:

$$\begin{aligned}
P_y = & HPH^T + E \left[\frac{D_{\bar{x}}h(D_{\bar{x}}^3h)^T}{3!} + \frac{D_{\bar{x}}^2h(D_{\bar{x}}^2h)^T}{2!2!} + \frac{D_{\bar{x}}^3h(D_{\bar{x}}h)^T}{3!} \right] + \\
& E \left(\frac{D_{\bar{x}}^2h}{2!} \right) E \left(\frac{D_{\bar{x}}^2h}{2!} \right)^T + \dots
\end{aligned} \quad (14.23)$$

This is the complete Taylor series expansion for the covariance of a nonlinear transformation.

In the EKF, we use only the first term of this expansion to approximate the covariance of the estimation error. For example, if the measurement $y = h(x) + v$ then we see from Equation (10.98) that the covariance of y is approximated as $P_y = HP_xH^T + R$, where H is the partial derivative of h with respect to x , and R is the covariance of v . Likewise, if the state propagates as $x_{k+1} = f(x_k) + w_k$ then we see from Equation (10.100) that the covariance of x is approximately updated as $P_k^- = FP_{k-1}^+F^T + Q$, where F is the partial derivative of $f(x)$ with respect to x , and Q is the covariance of w_k . However, these covariance approximations can result in significant errors if the underlying functions $h(x)$ and $f(x)$ are highly nonlinear.

For example, consider the nonlinear transformation introduced at the beginning of this section. A linear covariance approximation would indicate that $P_y \approx HP_xH^T$, where H and P_x are given as

$$\begin{aligned}
H &= \left. \frac{\partial h}{\partial x} \right|_{x=\bar{x}} \\
&= \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}_{x=\bar{x}} \\
&= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \\
P_x &= E \left(\begin{bmatrix} r - \bar{r} \\ \theta - \bar{\theta} \end{bmatrix} \begin{bmatrix} \dots \end{bmatrix}^T \right) \\
&= \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\theta^2 \end{bmatrix}
\end{aligned} \quad (14.24)$$

This gives P_y as follows.

$$\begin{aligned}
P_y &\approx HP_xH^T \\
&= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\theta^2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} \sigma_\theta^2 & 0 \\ 0 & \sigma_r^2 \end{bmatrix}
\end{aligned} \quad (14.25)$$

This is an approximation of P_y . However, a more rigorous analysis of P_y can be conducted using Equations (14.1), (14.7), and (14.10):

$$\begin{aligned}
P_y &= E[(y - \bar{y})(y - \bar{y})^T] \\
&= E \left[\begin{pmatrix} r \cos \theta \\ r \sin \theta - (\sin \theta_m)/\theta_m \end{pmatrix} \begin{pmatrix} \dots \end{pmatrix}^T \right] \\
&= E \begin{bmatrix} r^2 \cos^2 \theta & r^2 \cos \theta \sin \theta - r \cos \theta (\sin \theta_m)/\theta_m \\ r^2 \cos \theta \sin \theta - r \cos \theta (\sin \theta_m)/\theta_m & (r \sin \theta - (\sin \theta_m)/\theta_m)^2 \end{bmatrix}
\end{aligned} \quad (14.26)$$

We again use our assumption that r and θ are independent, r is uniformly distributed with a mean of 1 and a standard deviation of σ_r , and $\theta = \pi/2 + \tilde{\theta}$, with $\tilde{\theta}$ uniformly distributed between $\pm\theta_m$. We can therefore compute

$$\begin{aligned}
 E(r^2) &= 1 + \sigma_r^2 \\
 E(\cos^2 \tilde{\theta}) &= \frac{1 - E(\cos 2\tilde{\theta})}{2} \\
 E(\cos 2\tilde{\theta}) &= \frac{\sin 2\theta_m}{2\theta_m} \\
 E(\sin \theta) &= E(\cos \tilde{\theta}) \\
 &= \frac{\sin \theta_m}{\theta_m}
 \end{aligned} \tag{14.27}$$

We can use these expressions in Equation (14.26) to compute

$$P_y = \begin{bmatrix} \frac{1}{2}(1 + \sigma_r^2)(1 - \sin 2\theta_m/2\theta_m) & 0 \\ 0 & \frac{1}{2}(1 + \sigma_r^2)(1 + \sin 2\theta_m/2\theta_m) - \sin^2 \theta_m/\theta_m^2 \end{bmatrix} \tag{14.28}$$

This matrix defines a two-dimensional ellipse, where $P_y(1,1)$ specifies the square of the y_1 axis length, and $P_y(2,2)$ specifies the square of the y_2 axis length. Figure 14.2 shows the linearized covariance defined by Equation (14.25), and the exact covariance defined by Equation (14.28). The linearized covariance is centered at the linearized mean, and the exact covariance is centered around at the exact mean. It can be seen that the linearized covariance is not a very good approximation to the exact covariance, at least not in the y_2 direction.

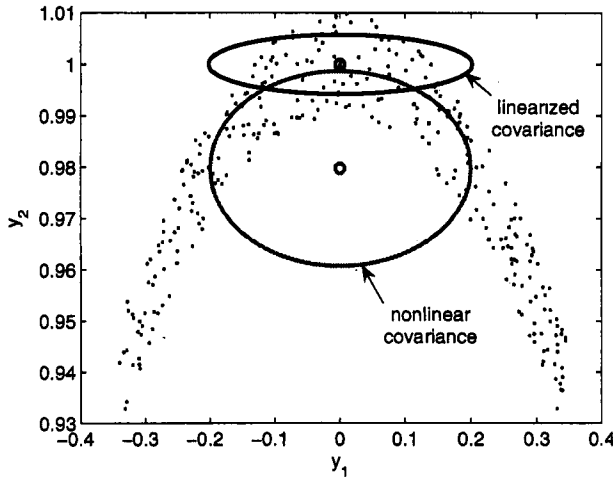


Figure 14.2 Linearized and nonlinear mean and covariance of 300 randomly generated points with $\tilde{r}$ uniformly distributed between ± 0.01 and $\tilde{\theta}$ uniformly distributed between ± 0.35 radians.

This is not a Kalman filtering example. But since the EKF uses first-order linearization to update the covariance of the state, this example shows the kind of error that can creep into the EKF when it is applied to a nonlinear system.